

Raghu Ram Komara

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PROFESSIONAL SUMMARY

Research Software Engineer with a Master's in Artificial Intelligence and Business Analytics from the University of South Florida, experienced in Python engineering, machine learning workflows, healthcare AI, and enterprise software development. Skilled in building research software, REST APIs, data pipelines, Docker-based environments, cloud deployments, databases, and reproducible ML experimentation workflows for healthcare, imaging, and clinical data use cases. Background includes distributed AI systems, full-stack healthcare applications, technical documentation, and collaborative biomedical research supporting privacy-preserving AI.

EDUCATION

Master of Science - Artificial Intelligence And Business Analytics GPA: 3.89/4.00
University of South Florida, Tampa, FL Aug 2024 - May 2026
Relevant Coursework: Machine Learning, Deep Learning, Application Development for Analytics, Cloud Solutions Architect, Big Data for Business, Advanced Database Management System, Foundations of Business Statistics

RESEARCH & PUBLICATIONS

- **Presented Research:** “Distinguishing Medical Diversity from Byzantine Attacks: Client-Adaptive Anomaly Detection for Healthcare Federated Learning” accepted as a Research-in-Progress paper at the Workshop on Information Technology and Systems (WITS), Nashville, USA, 2025.

SKILLS

- **Programming & Software Engineering:** Python, R, JavaScript, TypeScript, SQL, Java, C#, OOP, Data Structures & Algorithms, REST APIs, JSON, API Integration, Software Development Lifecycle (SDLC), Testing, Debugging
- **Research Software & ML Pipelines:** Machine Learning Pipelines, Analysis Pipeline Development, Reproducible Research Workflows, Experiment Tracking, Model Evaluation, Feature Engineering, Data Preprocessing, Documentation, User Support
- **AI & ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Federated Learning, NLP, LoRA, LangChain, LangGraph, RAG Pipelines, Agentic AI, LLM Evaluation
- **Data Systems & Databases:** PostgreSQL, MySQL, Snowflake, BigQuery, MongoDB, Redis, SQL/NoSQL, Database Design, Data Modeling, Query Optimization, Data Integration, Data Validation
- **Cloud, DevOps & Infrastructure:** Docker, Kubernetes, AWS, Azure, GCP, Linux, Virtual Machines, CI/CD Pipelines, Git, GitHub Actions, Jenkins, Containerized Deployment, Distributed Computing
- **Healthcare & Biomedical AI:** Healthcare AI, Biomedical Research Workflows, Clinical Data Workflows, Medical Imaging Data, Oncology-Related Data, HIPAA-Conscious Development, Data Integrity, Secure Data Handling

WORK EXPERIENCE

AI Engineer Intern, PODS Enterprises, LLC Tampa, FL, Jun 2026 – Present

- Developed Python-based AI software using Claude Enterprise, LangChain, and LangGraph to build production-ready LLM workflows with tool calling, structured outputs, error handling, and workflow orchestration.
- Built retrieval-augmented generation (RAG) systems integrating operational data sources through document ingestion, embeddings, vector search, reranking, grounding, and evaluation to deliver reliable AI-assisted information retrieval.
- Designed agentic AI software for route optimization by implementing planning workflows, state management, memory, and tool orchestration to automate operational decision-making and resource allocation.

Graduate Research & Teaching Assistant, University of South Florida Tampa, FL, May 2025 – May 2026

- Developed research software and distributed machine learning pipelines using Python, PyTorch, Flower, and Docker to support machine learning analyses of healthcare, medical imaging, and oncology-related datasets.
- Built software tools for data preprocessing, integration, feature engineering, model evaluation, visualization, and experiment tracking, enabling reproducible computational workflows for biomedical AI research.
- Maintained structured healthcare and medical imaging datasets by developing preprocessing, validation, and data organization workflows suitable for scalable machine learning experimentation and transparent research.
- Integrated machine learning frameworks, experiment tracking platforms (MLflow, Weights & Biases), and visualization tools to support reproducible research, software validation, and analysis of experimental results.
- Authored technical documentation, computational workflows, and literature reviews while supporting researchers and students in utilizing research software, reproducing experiments, and adopting AI-assisted development practices.
- Co-authored healthcare AI research on privacy-preserving federated learning by designing distributed experimentation environments and contributing to research presented at WITS 2025, with manuscript preparation underway for IEEE Journal of Biomedical and Health Informatics (J-BHI).

Full Stack Engineer, University of South Florida (Client: Evara Health) Tampa, FL, Oct 2025 – Jan 2026

- Designed and developed mobile and web-based healthcare applications using React Native, Python, FastAPI, and REST APIs to support real-time patient-provider communication and clinical workflows.
- Developed secure backend services and API integrations connecting healthcare applications with structured clinical data workflows, ensuring reliable processing, data integrity, and scalable application performance.
- Integrated Azure Cognitive Services and deployed Docker-based applications on Azure, supporting AI-powered transcription, multilingual communication, integration testing, and HIPAA-conscious software delivery.

Software Engineer, Accenture Hyderabad, India, Jan 2023 – Aug 2024

- Developed scalable enterprise web applications and backend services using React.js, TypeScript, Node.js, Python, FastAPI, and REST APIs to support secure, distributed business applications.
- Designed and optimized PostgreSQL and MySQL data models using indexing, normalization, query optimization, and data validation techniques to improve application performance, reliability, and data integrity.
- Built backend APIs, integration services, and server-side automation supporting secure data exchange, workflow automation, cross-system communication, testing, and application maintenance.
- Applied structured software engineering practices throughout the SDLC, including requirements analysis, system design, code reviews, testing, CI/CD, release management, Agile delivery, and production support using AWS, Git, Jenkins, and automated deployment pipelines.

PROJECTS

Federated Learning for Secure Intrusion Detection (FL-IDS): Built a Python, PyTorch, Flower, and Docker-based distributed ML platform for privacy-preserving model training and analysis across decentralized security datasets.

AI Cybersecurity SOC Agent: Developed FastAPI and Elastic Stack-based research software to integrate log data, automate analysis workflows, and support ML-assisted cybersecurity investigations.

Radiology Report to Structured Clinical Data: Built a biomedical NLP pipeline using Hugging Face Transformers and LoRA to convert radiology reports into structured clinical data for healthcare data integration and ML workflows.